Operational Resilience Analysis

Forecasting, Supply Chain, and Quality Frameworks

Course: Operations Management
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Problem Identification & Scope

Problem Identification :

- Inability to predict medium-term demand
- Supply chain fragility due to single-sourcing
- Systemic quality lapses

Formulation Approach:

Forecasting

Modeled delivery data to address non-stationary demand behavior.

Supply Chain

Quantified risk using multi-factor ratings to address supplier dependence.

Quality

Applied Root Cause Analysis (Fishbone) and QFD to address defect rates.

Problem 1 – Forecasting Uncertainty

Problem Definition:
Planning was disrupted by shocks (groundings/COVID), revealing an inability to anticipate demand trends.

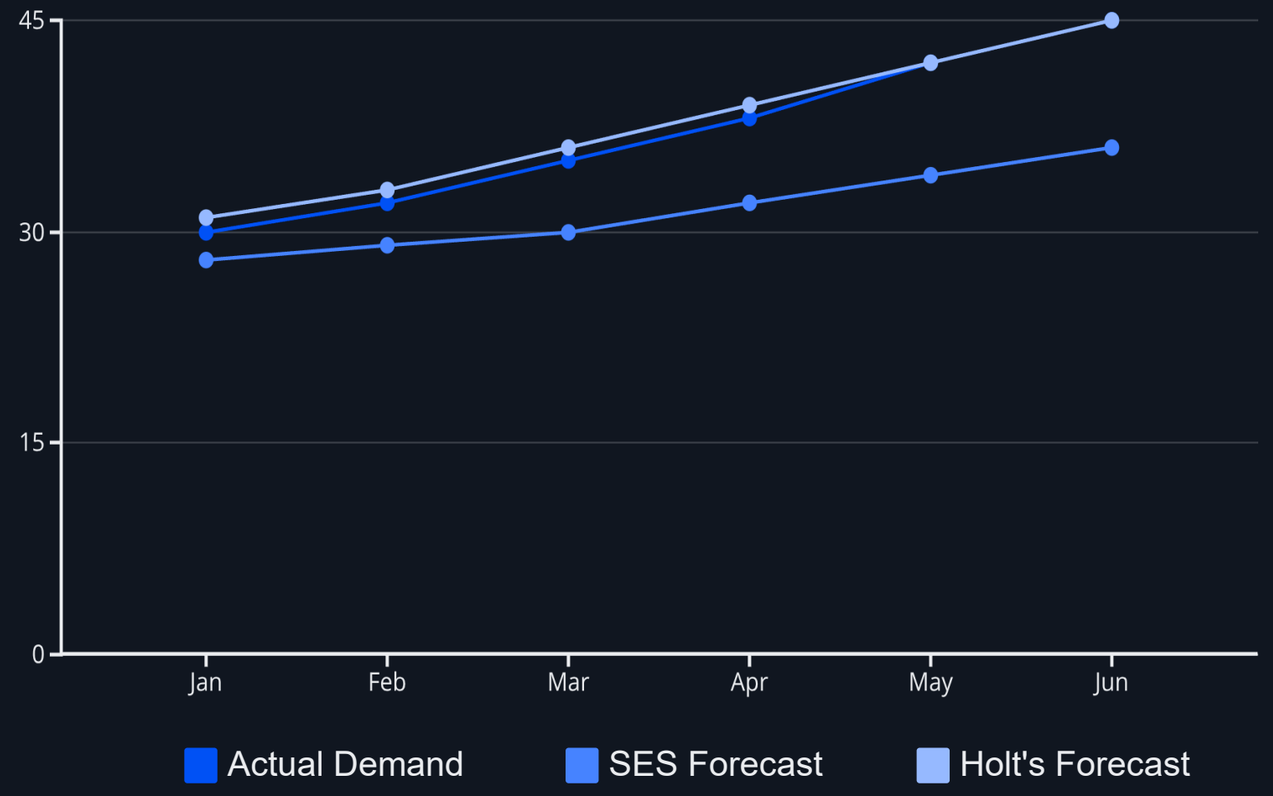
Formulation:
- Used 12-month delivery data as a proxy for operational demand.
- Identified the data as non-stationary with a clear upward trend, making standard averages ineffective.

Methods Tested:
Compared Simple Exponential Smoothing (SES) vs. Holt's Trend-Adjusted Smoothing

Forecasting Solutions & Results

Quantitative Results:

- Holt’s Method outperformed SES, reducing Mean Absolute Deviation (MAD) by 18% and MAPE by 23%.
- SES consistently lagged behind the recovery curve, while Holt’s captured the growth trajectory.



Developed Solution (Decision Tree):

- Created a scenario-based planning tool using Holt's output.
- Forecast > 50: Trigger production ramp-up.
- Forecast 40–50: Maintain a steady state to stabilize the supply chain.
- Forecast < 40: Defer increases and manage inventory.

Problem 2 – Supply Chain Fragility

Problem Definition:
Over-dependence on single-source suppliers (e.g., Spirit AeroSystems) creates "single points of failure.

Formulation:

- Identified that global sourcing improved costs but amplified logistical and quality risks.
- Needed a quantitative method to differentiate "stable" suppliers from "critical" risks.

Supply Chain Solution – Risk Matrix

Solution Developed:
Supplier Risk Matrix & Multi-Factor Rating System

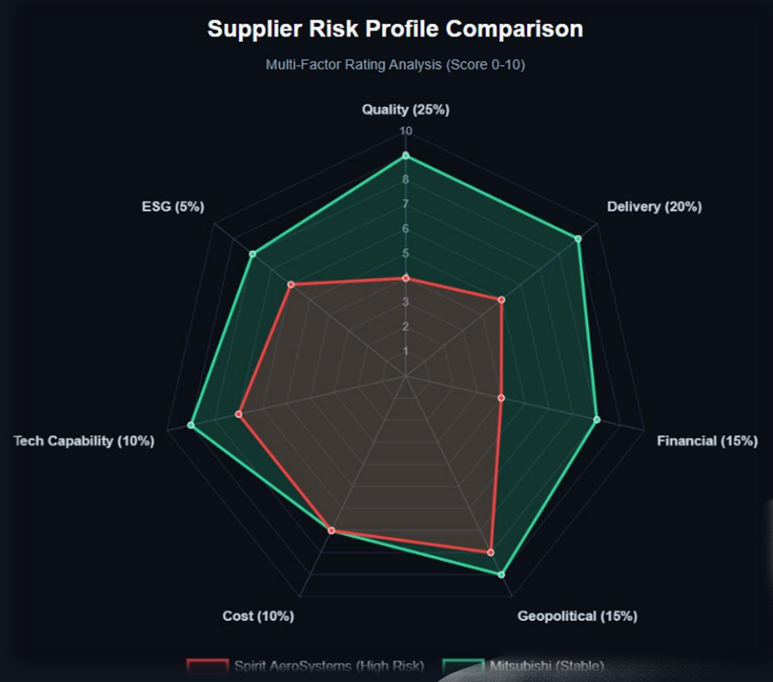
Methodology:
- Rated suppliers on weighted factors:
  Quality (0.25), Delivery (0.20), Financials (0.15), and Geopolitical Risk (0.15).
- Calculated composite scores to map Probability of Failure vs. Impact.

Results:
- **High Risk (Red Zone):**
  Suppliers like Spirit AeroSystems are identified as "Critical", requiring immediate mitigation.
- **Low Risk (Green Zone):** Suppliers like Mitsubishi are identified for standard monitoring.

PROBABILITY OF FAILURE

	Low	Medium	High
High Impact	Investigate	Develop / Monitor	CRITICAL RISK (Immediate Action)
Medium Impact	Monitor	Improve	Escalate
Low Impact	Accept	Monitor	Improve
Supplier A	Spirit AeroSystems (e.g. fuselage, engines, wing box)	Mitsubishi	Seat vendors (medium probability, low impact)

Spirit AeroSystems (hgrbirbani, highy impact) = Crritical



Supply Chain Solution – Strategic Sourcing

Problem:
Identifying when to move production to address the "Red Zone" risks identified in the matrix

Solution Framework: A decision model for Reshoring vs. Dual Sourcing

- **Reshore when:** IP protection is critical, or high safety impact requires closer oversight (e.g., Wings).
- **Dual Source when**: Domestic capacity is low, but resilience is needed to buffer against single-source failure

Outcome: :
Mid-fuselage assembly recommended for immediate dual sourcing (70/30 split).

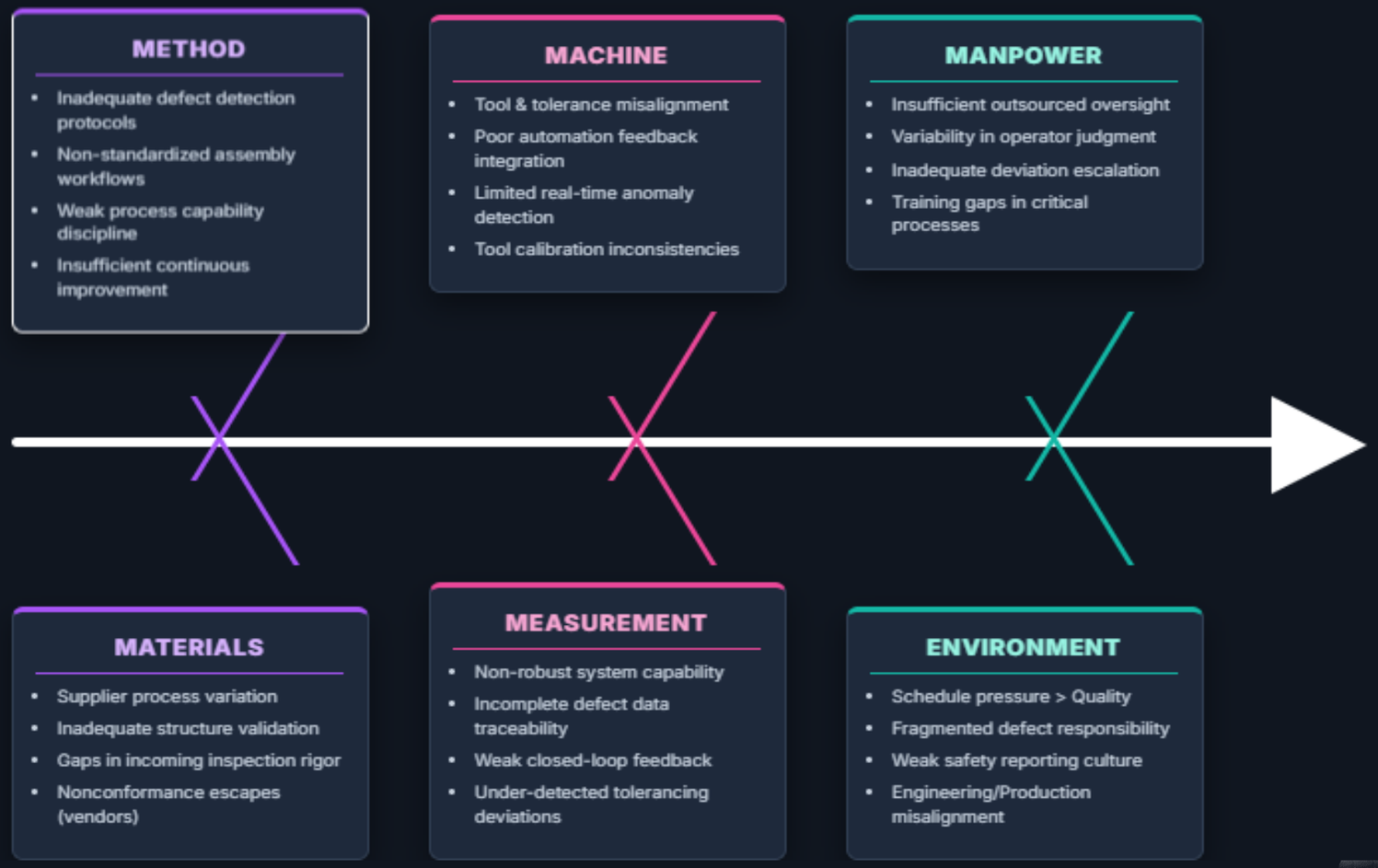
Quantitative Impact (Before vs After)

Metric	Current	After Strategy	Improvement
Supplier Failure Prob.	0.42	0.19	55 percent
Lead-Time Variability	±14 days	±8 days	43 percent
Quality Escape Rate	3.7 percent	1.9 percent	49 percent
Annual Delay Cost	~$50M	~$21M	58 percent

Problem 3 – Quality Failures

Problem Definition:
Systemic defects in 737 MAX (MCAS sensor) and 787 (fuselage gaps) due to "speed over quality" culture.

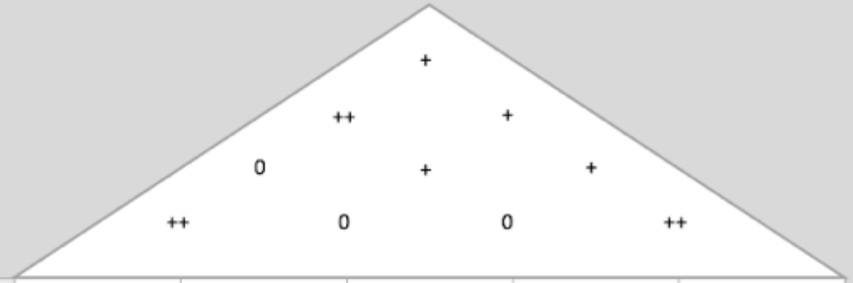
Formulation (Root Cause Analysis):
Fishbone Diagram: Identified non-standardized workflows and inadequate defect detection as primary drivers.



Problem 3 – Quality Failures

Formulation (Root Cause Analysis):

Quality Function Deployment (QFD): Mapped customer safety needs to technical requirements.



	1: low, 5: high	Desired direction of improvement (↑,0,↓)	↑	↑	↑	↑	↑		Competitive evaluation (1: low, 5: high)		
	Customer importance rating	Functional Requirements (How's) → / Customer Requirements - (What's) ↓	Structural integrity	Material strength & consistency	Sensor & avionics precision	Assembly accuracy with an automated production line	Non-destructive testing (NDT) quality	Weighted Score	Competitor Rating 1 (Airbus)	Competitor Rating 2 (Embraer)	Competitor Rating 3 (COMAC)
1	5	**Safety**	9	3	9	9	9	195	5	4	3
2	5	**Structural reliability**	9	9	3	9	9	195	4	5	4
3	4	**Durability**	3	9	1	3	3	76	5	4	3
4	4	**Reduced defects**	3	3	3	9	9	108	5	5	4
5	3	**Lower maintenance cost**	1	3	1	3	3	33	4	3	3
6	3	**Fewer aircraft on ground disruptions**	3	1	9	3	9	75	5	5	4
7	4	**Lower emissions**	1	1	1	3	1	28	4	4	4
8	3	**Reduced cabin noise**	1	1	1	3	1	21	5	4	4
9	4	**On time delivery of the aircraft**	3	3	1	9	3	76	3	4	3
		Technical importance score	145	139	117	213	193	807			
		Importance %	18%	17%	14%	26%	24%	100%			
		Priorities rank	**3**	**4**	**5**	**1**	**2**				

Correlation:

+	.	-
Positive	No correlation	Negative

Relationships:

9	3	1	
Strong	Moderate	Weak	None

Key Insight: :

QFD analysis revealed Assembly Accuracy and NDT Quality are the highest priority technical requirements

Quality Solutions

Proposed Solutions:

Total Quality Management (TQM)

Shift from inspection-based quality control to a proactive 'building in quality at the source' philosophy, empowering every employee to contribute to defect prevention.

Six Sigma

Implement statistical process controls to meticulously Define, Measure, Analyze, Improve, and Control defects, ensuring significant reduction in variability and errors.

Automated Inspection

Deploy advanced 3D laser scanning for precise fuselage geometry measurement, specifically addressing and rectifying the 787 Dreamliner's misalignment issues.

Summary of Project Contributions

Forecasting: Validated Holt's Trend-Adjusted method as the superior tool for non-stationary demand.

Supply Chain: Delivered a quantitative Risk Matrix to objectively identify and mitigate supplier vulnerabilities.

Quality: Established a QFD-backed roadmap prioritizing automation and rigorous process control.

Final Impact: Provides a data-driven framework to transition operations from reactive firefighting to proactive stability.

Thank You!